

Question Number	Answer	Additional guidance	Mark
7(a)(i)	A - Metaphase I The only correct answer is A <i>B is not correct because independent assortment does not occur in metaphase II</i> <i>C is not correct because independent assortment does not occur in telophase I</i> <i>D is not correct because independent assortment does not occur in telophase II</i>		(1)

Question Number	Answer	Additional guidance	Mark
7(a)(ii)	C - Prophase I The only correct answer is C <i>A is not correct because crossing over does not occur in metaphase I</i> <i>B is not correct because crossing over does not occur in metaphase II</i> <i>D is not correct because crossing over does not occur in prophase II</i>		(1)

Question Number	Answer	Additional guidance	Mark
7(b)	An explanation that includes three of the following points: • (due to) linkage (1) • A and B are on same chromosome (1) • {genes/ alleles / AB / ab / 48%} are not separated due to crossing over / {AB / ab} are inherited together (1) • {Ab / aB / 2%} formed by crossing over (1)	ACCEPT 25% due to A and B on separate chromosomes ACCEPT the closer together two loci are the smaller the chance of separation (by recombination) / ORA ACCEPT {Ab / aB} are recombinants	(3)

Question Number	Answer	Additional guidance	Mark
7(c)	A description that includes the following points: • reference to post transcriptional changes / (pre-)RNA splicing (1) • {introns / P, R, T, V and X} removed by enzymes (1) • rearrangement of {exons / Q, S, U and W} / removal of some exons (1) • credit two different permutations of exon order given (1) • (therefore) a different {primary sequence / sequence of amino acids / polypeptide} (1)	ACCEPT spliceosomes ACCEPT not all exons used ACCEPT different sequence of amino acids can result in differences in {folding / bonding} giving a different protein (3D) shape	(5)

Question Number	Answer	Additional guidance	Mark
7(d)	An answer that includes the following points: • (activity of) gene T decreases (during development) and gene U increases (1) • (because) gene T is switched off (1) • {product of gene T not needed / (gene T) protein not produced} once blastocyst stage reached (1) • product of gene U is needed at higher levels after the {8-cell / morula} stage (1)	ACCEPT pieced together ACCEPT description ACCEPT product of gene T not needed after 8-cell stage reached ACCEPT gene U is involved in specialisation of the cells / correct description of role of gene U protein	(3)